Math 161
Problem set 1 solutions

Justin Campbell

1. (a)

P	T	T	T	T	F	F	F	F
Q	T	T	F	F	T	T	F	F
R	T	F	T	F	T	F	T	F
not P	F	F	F	F	T	T	T	T
Q and R	T	F	F	F	T	F	F	F
(not P) $\implies$ (Q and R)	T	T	T	T	T	F	F	F

   (b)

P	T	T	F	F
Q	T	F	T	F
$P \implies Q$	T	F	T	T
not ($P \implies Q$)	F	T	F	F
not Q	F	T	F	T
P and not Q	F	T	F	F
not ($P \implies Q$) $\implies$ (P and not Q)	T	T	T	T

   (c)

P	T	T	F	F
Q	T	F	T	F
(P and not Q) $\implies$ not ($P \implies Q$)	T	T	T	T

   Since both "not ($P \implies Q$) $\implies$ (P and not Q)" and "(P and not Q) $\implies$ not ($P \implies Q$)" are always true, the statements "not ($P \implies Q$)" and "P and not Q" are logically equivalent. In fact, we can already see this from the fact that "not ($P \implies Q$)" and "P and not Q" have the same truth value for all truth values of P and Q.

2. The following solutions are meant to be a sampling of different ways to phrase these arguments: they are not necessarily the most efficient proofs (nor the least efficient). Your proofs may look quite different and still be correct.

   (a) Suppose that $x \in (A \cap B) \cup (A \cap (X \setminus B))$. By the definition of union, this means that either $x \in A \cap B$ or $x \in A \cap (X \setminus B)$. By the definition of intersection, the former means that $x \in A$ and $x \in B$, while the latter means that $x \in A$ and $x \in X \setminus B$. In either case we have $x \in A$, so we have shown that $(A \cap B) \cup (A \cap (X \setminus B)) \subset A$.

   For the other inclusion, assume that $x \in A$. Either $x \in B$ or $x \notin B$, and since $x \in A \subset X$, the latter case means that $x \in X \setminus B$ by the definition of complement. Thus, by the definition of intersection, either $x \in A \cap B$ or $x \in A \cap (X \setminus B)$. Applying the definition of union, we see that $x \in (A \cap B) \cup (A \cap (X \setminus B))$ and hence $A \subset (A \cap B) \cup (A \cap (X \setminus B))$.

   Combining these two inclusions yields the desired equality of sets $(A \cap B) \cup (A \cap (X \setminus B)) = A$.

(b) By part (d) of this problem, we have

$$(A \cup B) \setminus (A \cap B) = ((A \cup B) \setminus A) \cup ((A \cup B) \setminus B).$$

We claim that $(A \cup B) \setminus B = A \setminus B$. Namely, by the definition of union and complement, the condition that $x \in (A \cup B) \setminus B$ means

$$(x \in A \text{ or } x \in B) \text{ and not } x \in B.$$

But this is equivalent to ($x \in A$ and not $x \in B$), as we can see from the truth table

P	T	T	F	F
Q	T	F	T	F
P or Q	T	T	T	F
not Q	F	T	F	T
(P or Q) and not Q	F	T	F	F
P and not Q	F	T	F	F

By the definition of complement, the condition $x \in A$ and not $x \in B$ means that $x \in A \setminus B$. So we have shown that $(A \cup B) \setminus B = A \setminus B$. By symmetry (exchanging A and B), we see that $(A \cup B) \setminus A = B \setminus A$. Thus we have

$$((A \cup B) \setminus A) \cup ((A \cup B) \setminus B) = (A \setminus B) \cup (B \setminus A),$$

and combining with the equation above we obtain $(A \cup B) \setminus (A \cap B) = (A \setminus B) \cup (B \setminus A)$ as needed.

(c) We show that each subset is contained in the other.

Suppose that $x \in A \cup (B \cap C)$. By the definition of union, this means that $x \in A$ or $x \in B \cap C$. By the definition of intersection, the latter means that $x \in B$ and $x \in C$. Thus $x \in A$ or $x \in B$, and also $x \in A$ or $x \in C$. Applying the definitions of intersection and union again, we obtain $x \in (A \cup B) \cap (A \cup C)$. So we have shown that

$$A \cup (B \cap C) \subset (A \cup B) \cap (A \cup C).$$

Now suppose that $x \in (A \cup B) \cap (A \cup C)$. By the definition of intersection, this means that $x \in A \cup B$ and $x \in A \cup C$. By the definition of union, the former means that $x \in A$ or $x \in B$, while the latter means that $x \in A$ or $x \in C$. If $x \notin A$ then the first statement implies that $x \in B$, while the second statement implies that $x \in C$. Thus, by the definition of intersection we have shown that either $x \in A$ or $x \in B \cap C$, which by the definition of union means that $x \in A \cup (B \cap C)$. We have therefore proved that

$$(A \cup B) \cap (A \cup C) \subset A \cup (B \cap C),$$

as desired.

(d) We will use the following two tautologies (also sometimes called De Morgan's law), which hold for any propositions P and Q:

$$\text{not } (P \text{ or } Q) \iff ((\text{not } P) \text{ and } (\text{not } Q))$$

and

$$\text{not } (P \text{ and } Q) \iff ((\text{not } P) \text{ or } (\text{not } Q)).$$

These can be read off of the truth table

P	T	T	F	F
Q	T	F	T	F
P or Q	T	T	T	F
P and Q	T	F	F	F
not P	F	F	T	T
not Q	F	T	F	T
not (P or Q)	F	F	F	T
(not P) and (not Q)	F	F	F	T
not (P and Q)	F	T	T	T
(not P) or (not Q)	F	T	T	T

By the definitions of complement and union, the condition $x \in A \setminus (B \cup C)$ means that

$$x \in A \text{ and not } (x \in B \text{ or } x \in C).$$

Applying the first of the above tautologies with P being $x \in B$ and Q being $x \in C$, this statement is equivalent to

$$x \in A \text{ and } x \notin B \text{ and } x \notin C.$$

This is equivalent to

$$(x \in A \text{ and } x \notin B) \text{ and } (x \in A \text{ and } x \notin B)$$

(because repeating $x \in A$ has no effect), which by the definitions of complement and intersection means that $x \in (A \setminus B) \cap (A \setminus C)$. Thus we have shown that

$$(x \in A \setminus (B \cup C)) \Longleftrightarrow (x \in (A \setminus B) \cap (A \setminus C)),$$

i.e. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

For the second equality: by the definitions of complement and intersection, the condition $x \in A \setminus (B \cap C)$ means that

$$x \in A \text{ and not } (x \in B \text{ and } x \in C).$$

Applying the second of the above tautologies with P being $x \in B$ and Q being $x \in C$, this statement is equivalent to

$$x \in A \text{ and } (x \notin B \text{ or } x \notin C).$$

By the definitions of intersection, complement, and union, this is equivalent to

$$x \in A \cap ((X \setminus B) \cup (X \setminus C)),$$

so we have shown that $A \setminus (B \cap C) = A \cap ((X \setminus B) \cup (X \setminus C))$. By the distributivity of intersection over union (shown in my example proof), we have

$$A \cap ((X \setminus B) \cup (X \setminus C)) = (A \cap (X \setminus B)) \cup (A \cap (X \setminus C)).$$

Note that $A \cap (X \setminus B) = A \setminus B$, since $x \in A \cap (X \setminus B)$ and $x \in A \setminus B$ both mean that $x \in A$ and $x \notin B$. Thus we have

$$(A \cap (X \setminus B)) \cup (A \cap (X \setminus C)) = (A \cap B) \cup (A \cap C),$$

which combined with the previous equalities yields the desired

$$A \setminus (B \cap C) = (A \cap B) \cup (A \cap C).$$

3. (a) This is false in general. For example, let $X = \{1\}$, $Y = \{1, 2\}$, $f : X \to Y$ given by $f(1) = 1$, and $B = \{2\}$. Then $f^{-1}(B) = \varnothing$, so

$$f(f^{-1}(B)) = f(\varnothing) = \varnothing.$$

But $B \neq \varnothing$ and therefore B is not contained in $f(f^{-1}(B))$.

(b) Choose $y \in f(f^{-1}(B))$. By the definition of image, there exists $x \in f^{-1}(B)$ such that $f(x) = y$. By the definition of preimage, we have $f(x) \in B$, so $y = f(x) \in B$. Thus $f(f^{-1}(B)) \subset B$ as desired.

(c) This is false in general. For example, let $X = \{1, 2\}$, $Y = \{1\}$, $f : X \to Y$ given by $f(1) = 1$ and $f(2) = 1$, and $A = \{1\}$. Then $f(A) = \{1\}$, but

$$f^{-1}(f(A)) = f^{-1}(\{1\}) = \{1, 2\} \neq \{1\} = A.$$

(d) Fix $y \in f(A) \cap B$. By the definition of intersection, this means that $y \in f(A)$ and $y \in B$. By the definition of image, the former means that there exists some $x \in A$ such that $y = f(x)$. Note that since $f(x) = y \in B$, by the definition of preimage we have $x \in f^{-1}(B)$. By the definition of intersection, we therefore have $x \in A \cap f^{-1}(B)$. Finally, the definition of image yields

$$y = f(x) \in f(A \cap f^{-1}(B)).$$

(e) Choose $y \in f(A \cap f^{-1}(B))$. By the definition of image, we have $y = f(x)$ for some $x \in A \cap f^{-1}(B)$. By the definition of intersection, it follows that $x \in A$ and $x \in f^{-1}(B)$. The latter means (by the definition of preimage) that $f(x) \in B$. Since $x \in A$, the definition of preimage implies that $f(x) \in f(A)$. Finally, the definition of intersection allows us to deduce that $y = f(x) \in f(A) \cap B$, as needed.

4. (a) The hypothesis that A and B are countably infinite means that there exist bijections $f : \mathbb{N} \xrightarrow{\sim} A$ and $g : \mathbb{N} \xrightarrow{\sim} B$. Consider the function $h : \mathbb{N} \to A \cup B$ defined by

$$h(n) := \begin{cases} f(\frac{n+1}{2}) & \text{for } n \text{ odd;} \\ g(\frac{n}{2}) & \text{for } n \text{ even.} \end{cases}$$

We claim that h is surjective. Namely, if $x \in A \cup B$, then either $x \in A$ or $x \in B$. If $x \in A$, then we have $f(n) = x$ for some $n \in \mathbb{N}$ by the surjectivity of f, so $h(2n-1) = f(n) = x$. Similarly, if $x \in B$ then $g(n) = x$ for some $n \in \mathbb{N}$ by the surjectivity of g, and then we have $h(2n) = g(n) = x$.

By my first hint, the existence of a surjection $h : \mathbb{N} \to A \cup B$ implies that $A \cup B$ is finite or countably infinite. Since $A \cup B$ contains the infinite subset A, it is infinite by (the contrapositive of) my first hint. Thus $A \cup B$ is countably infinite.

(b) Note that $X = A \cup (X \setminus A)$. Namely, we have $A \cup (X \setminus A) \subset X$ because $A, X \setminus A \subset X$, and if $x \in X$ then either $x \in A$ or $x \notin A$. The latter case means that $x \in X \setminus A$, so in either case $x \in A \cup (X \setminus A)$ and we obtain $X \subset A \cup (X \setminus A)$.

It follows that if $X \setminus A$ is countable, then X is countable by part (a), contrary to hypothesis. Therefore $X \setminus A$ is uncountable.

5. As suggested, we begin by proving that statements (b) and (c) hold for $X = \mathbb{N}$. For (b), the map $\mathbb{N} \to \mathbb{N}$ given by $n \mapsto n+1$ is injective, because if $m+1 = n+1$ for some $m, n \in \mathbb{N}$ then

$$m = (m+1) - 1 = (n+1) - 1 = n.$$

But it is not surjective, since if $n + 1 = 1$ then

$$n = (n+1) - 1 = 1 - 1 = 0 \notin \mathbb{N},$$

and hence 1 is not the range of this map.

As for (c), consider the map $\mathbb{N} \to \mathbb{N}$ which sends $1 \mapsto 1$ and $n \mapsto n-1$ for $n > 1$. It is surjective, since $n = (n+1) - 1$ for any $n \in \mathbb{N}$. But it is not injective, since it sends $1, 2 \mapsto 1$.

(a) $\implies$ (b): By my first hint, there exists an injection $g : \mathbb{N} \to X$ for any infinite set X. Let $h : \mathbb{N} \to \mathbb{N}$ be a map which is injective but not surjective, for instance the one we constructed above. Now we define a map $f : X \to X$ as follows. If $x \in g(\mathbb{N})$, so that $x = g(n)$ for some $n \in \mathbb{N}$, we put $f(x) := g(h(n))$. If $x \notin g(\mathbb{N})$, then define $f(x) := x$.

To see that f is injective, suppose that $x, y \in X$ are such that $f(x) = f(y)$. If $x, y \in g(\mathbb{N})$, say $x = g(m)$ and $y = g(n)$, then we have

$$g(h(m)) = f(x) = f(y) = g(h(n))$$

by the definition of f. Since g was assumed injective, this implies that $h(m) = h(n)$. We also chose h to be injective, so $m = n$ and hence

$$x = g(m) = g(n) = y.$$

Another possible case is $x, y \notin g(\mathbb{N})$, which immediately implies that

$$x = f(x) = g(y) = y.$$

If $x \in g(\mathbb{N})$, say $x = g(n)$, but $y \notin g(\mathbb{N})$, then we have

$$g(h(n)) = f(x) = f(y) = y.$$

But this means that $y \in g(\mathbb{N})$, contrary to hypothesis, so we have obtained a contradiction and this case is impossible. Exchanging x and y in the previous case shows that it is also impossible to have $x \notin g(\mathbb{N})$ and $y \in g(\mathbb{N})$ simultaneously. This completes the proof that f is injective.

It remains to prove that f is not surjective. We chose h to be non-surjective, so there exists $n \in \mathbb{N}$ such that $h(m) \neq n$ for any $m \in \mathbb{N}$. We claim that $g(n) \notin f(\mathbb{N})$, which will show that f is not surjective. Suppose for the sake of contradiction that there exists $x \in X$ such that $f(x) = g(n)$. If $x \in g(\mathbb{N})$, so that $x = g(m)$ for some $m \in \mathbb{N}$, then we have

$$g(n) = f(x) = g(h(m))$$

by the definition of f. Since g is injective this implies that $n = h(m)$, which contradicts our choice of n. If $x \notin g(\mathbb{N})$, then we have

$$x = f(x) = g(n) \in g(\mathbb{N})$$

by the definition of f, which is another contradiction. Thus there does not exist $x \in X$ such that $f(x) = g(n)$, i.e. $g(n) \notin f(X)$ as claimed.

(a) $\implies$ (c): Again we begin by choosing an injection $g : \mathbb{N} \to X$, which exists for any infinite set X. Let $h : \mathbb{N} \to \mathbb{N}$ be a map which is surjective but not injective, such as the example constructed above. We define $f : X \to X$ as in the previous implication: if $x \in g(\mathbb{N})$, i.e. $x = g(n)$ for some $n \in \mathbb{N}$, put $f(x) := g(h(n))$. If $x \notin g(\mathbb{N})$, then let $f(x) := x$.

Let us prove that f is surjective. Given $x \in X$, either $x \in g(\mathbb{N})$ or $x \notin g(\mathbb{N})$. In the first case we have $x = g(n)$ for some $n \in \mathbb{N}$, and since h was chosen surjective, there exists $m \in \mathbb{N}$ such that $n = h(m)$. Thus

$$x = g(n) = g(h(m)) = f(g(m))$$

by the definition of f. In the case $x \notin g(\mathbb{N})$, we have $x = f(x)$ by the definition of f. Thus in either case $x \in f(X)$, and we have shown that f is surjective.

Finally, we show that f is not injective. Since h is not injective, there exist $m, n \in \mathbb{N}$ such that $h(m) = h(n)$ but $m \neq n$. Then

$$f(g(m)) = g(h(m)) = g(h(n)) = f(g(n))$$

by the definition of f. If $g(m) = g(n)$ then $m = n$ by the injectivity of g, but this contradicts our choice of m and n, so $g(m) \neq g(n)$ and we have shown that f is not injective.

(b) $\implies$ (a): The contrapositive of this statement says that if X is finite, then there does not exist a map $f : X \to X$ which is injective but not surjective. This was part of my second hint.

(c) $\implies$ (a): Again my second hint implies the contrapositive, which says that if X is finite then any surjective map $f : X \to X$ is injective.